



Multicore and Manycore Architecture Basics

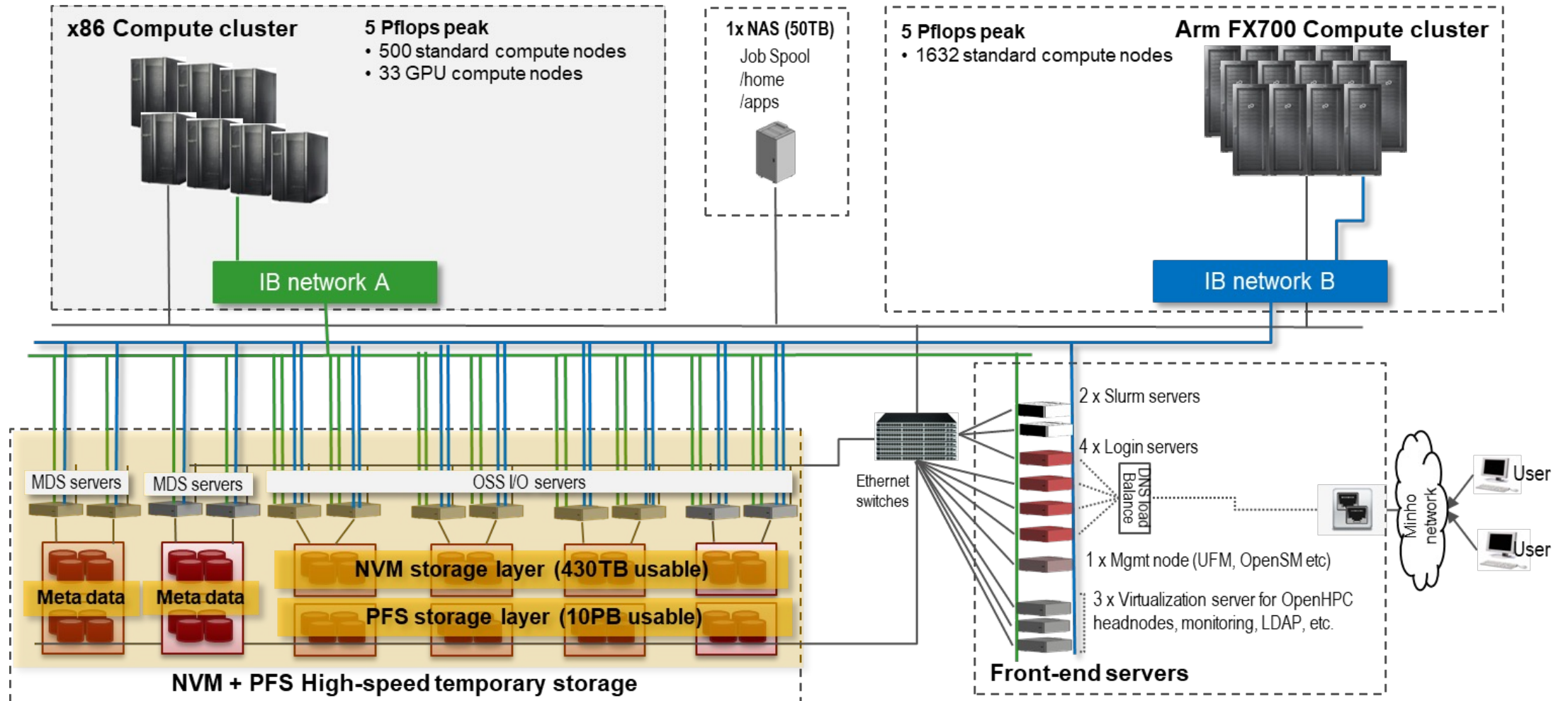
2025

André Pereira

Deucalion – Where a `clear` doesn't take 20 secs



Application Profiling

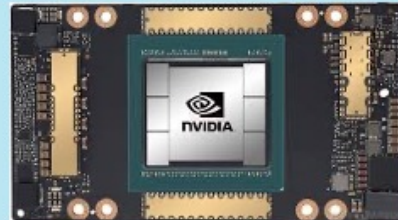




Compute nodes – 1632
Cores Number – 78,336
Memory Capacity – 52 TB
Rpeak – 5.013 PFlops



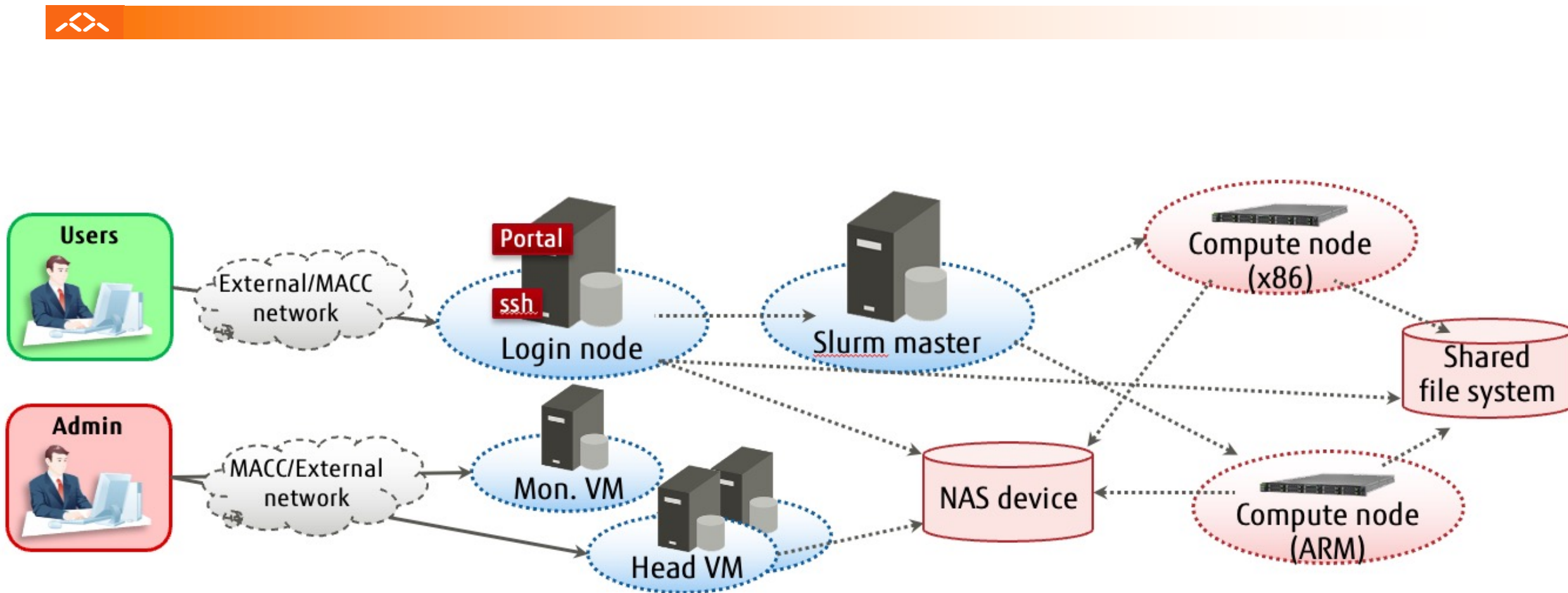
Compute nodes – 500
Cores Number – 64,000
Memory Capacity – 128 TB
Rpeak – 2.304 PFlops



Compute nodes – 33
CPU Cores Number – 4,224
Memory Capacity – 16 TB
GPU Memory – 8 TB
Rpeak CPU – 152,064 GFlops
Rpeak GPU – 2.572 PFlops

10 PFlops

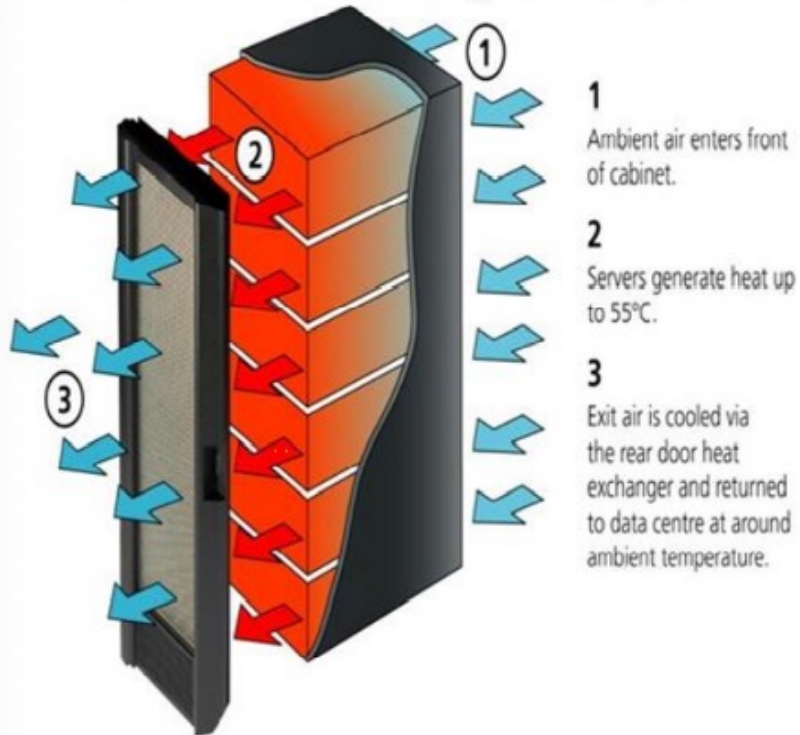
Application Profiling



A Bit of General Knowledge - Cooling



Rear Door Heat Exchanger concept

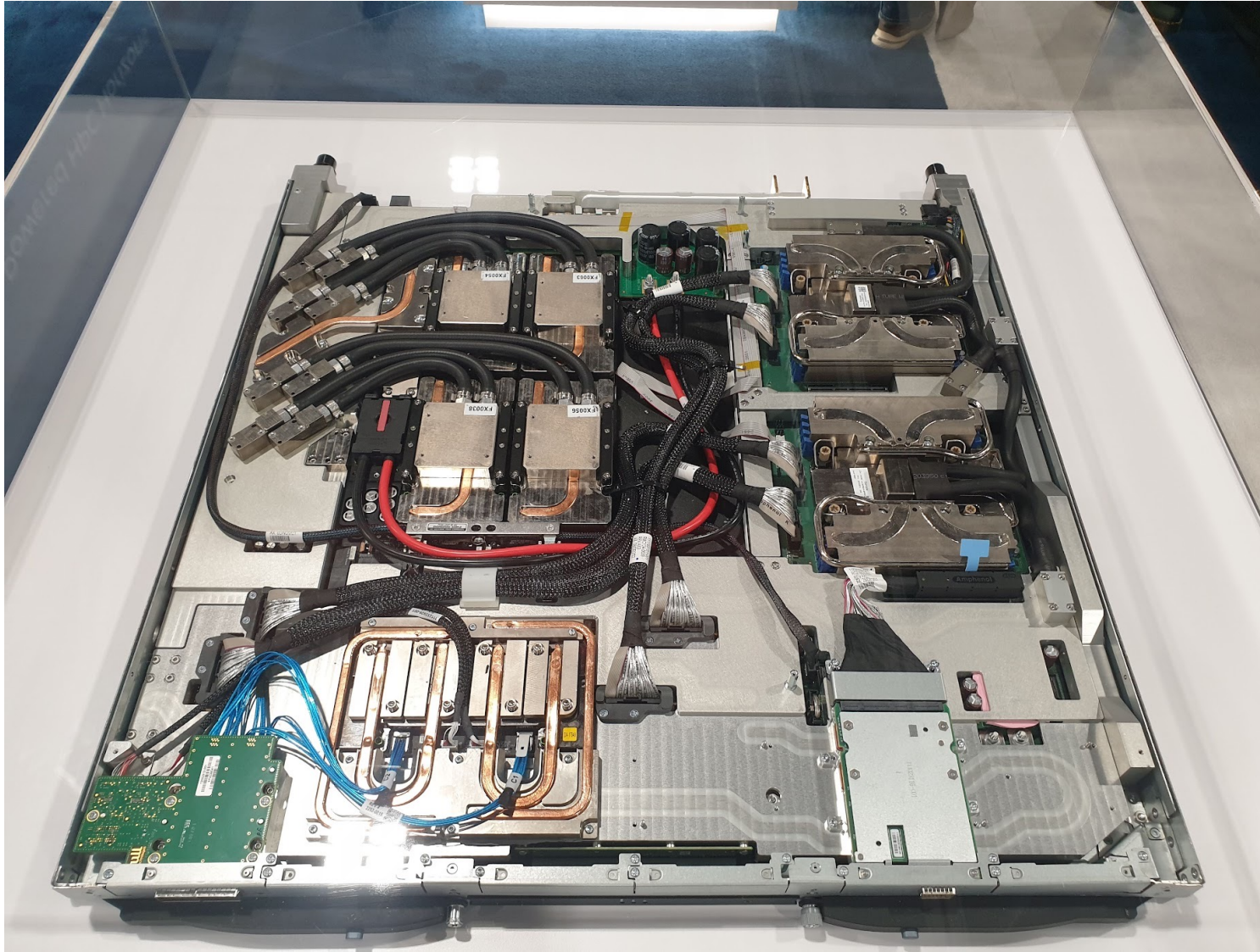


DCD-50 unit
Mounted on a rack



Cooling Capacity per Rack – 50kW
Server inlet temperature – 30 °C
Water in: 18°C
Water out: 26°C

Cooling – Denser Servers Coming Out

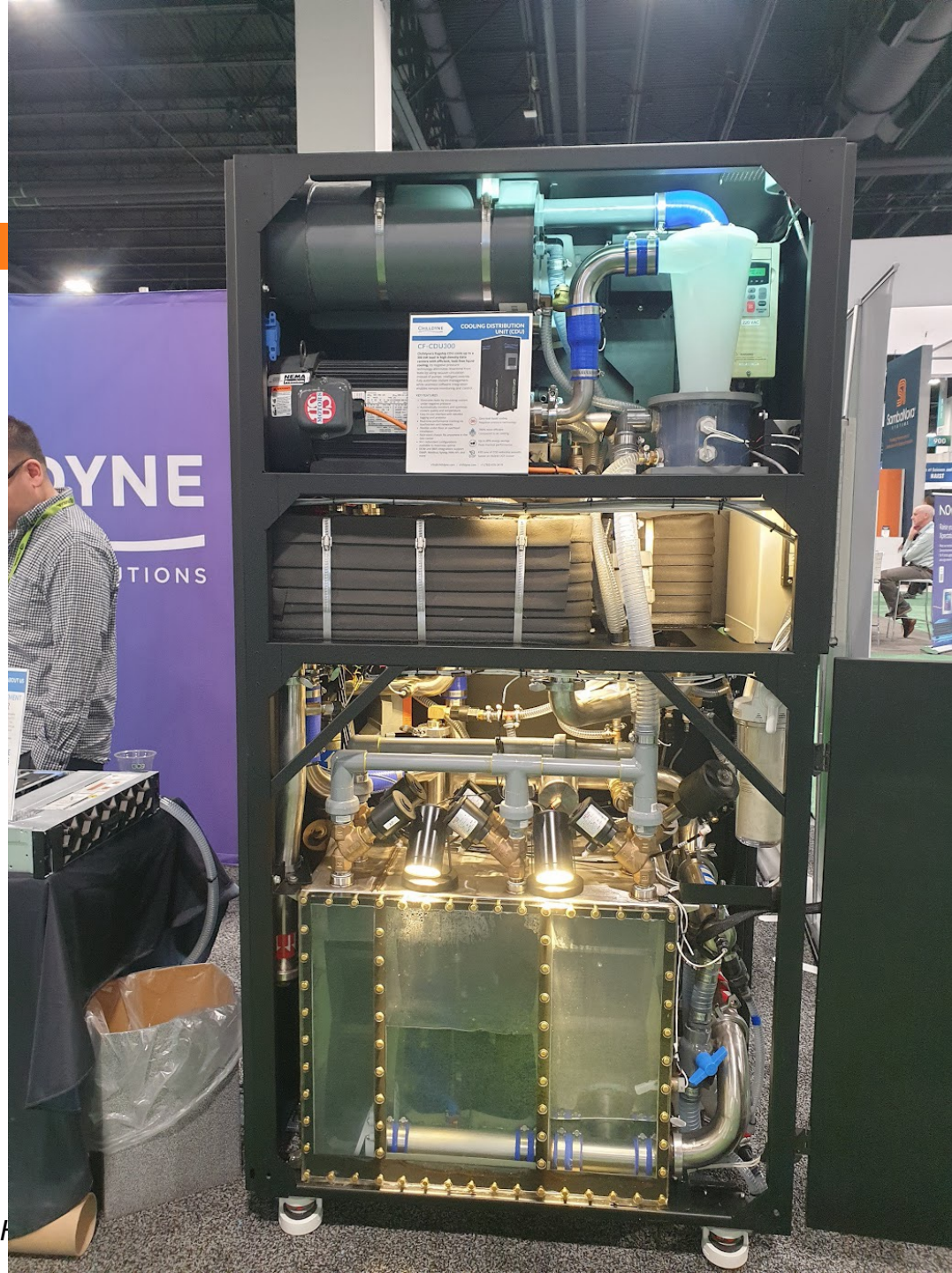


Cooling – Denser Servers Coming Out





Cooling – Water Cool Everything!



Cooling – The Layout is Crucial

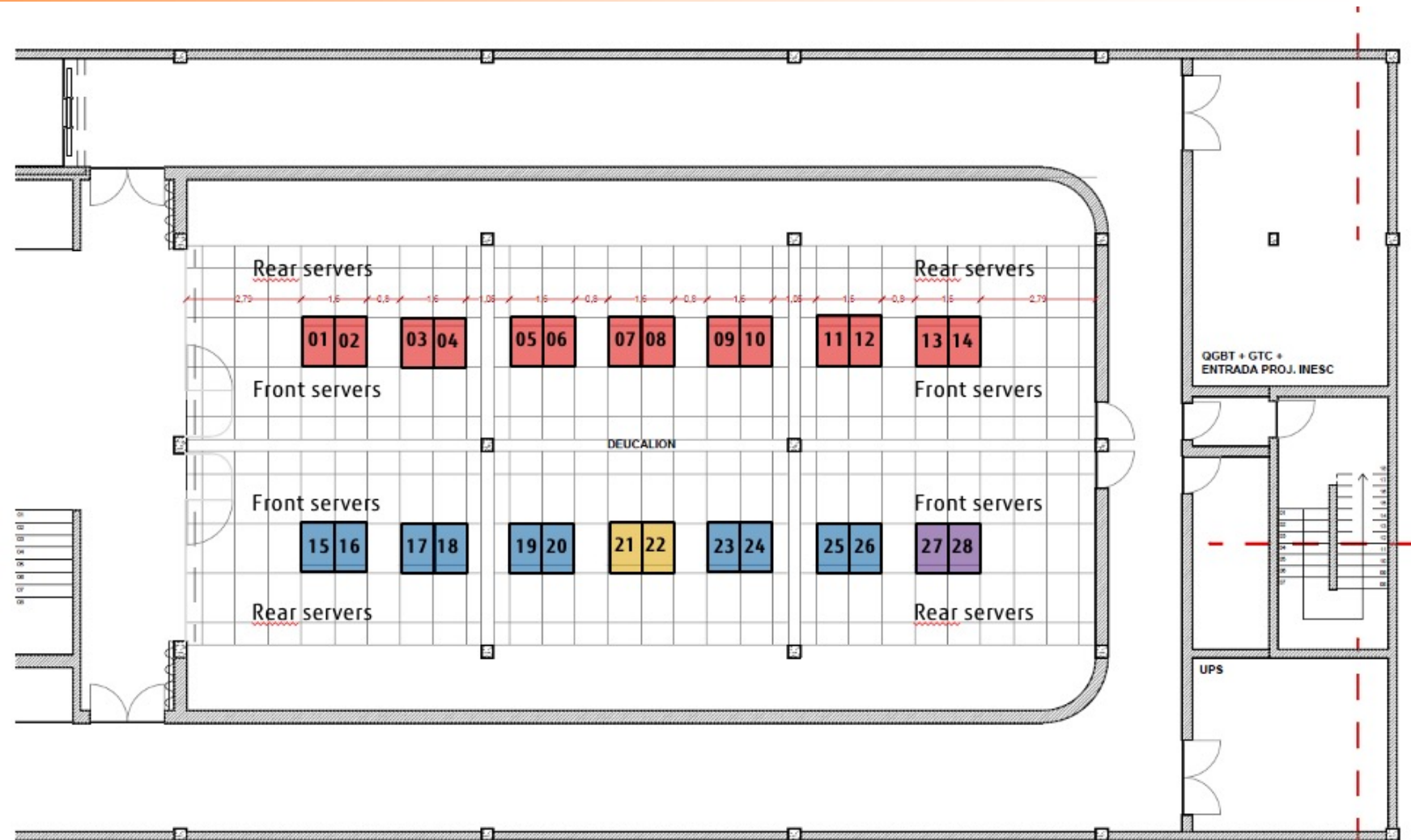


RACK ARM
CLUSTER
(RA)

RACK X86
CLUSTER
(RX)

RACK
STORAGE
(RS)

RACK
COMMON
(RC)



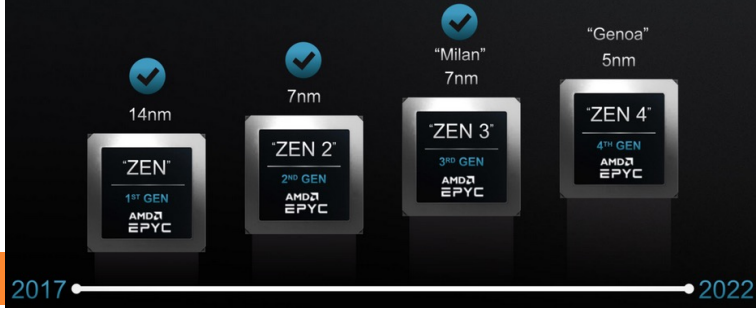


From Multicore to Manycore

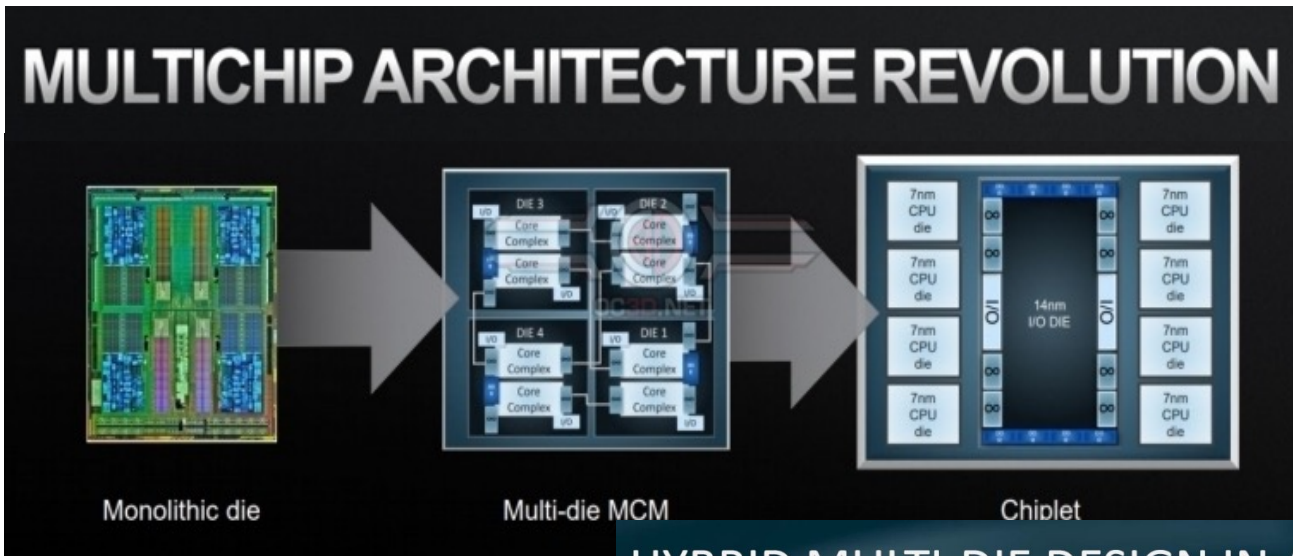


Tips to fit many cores into a single chip

- lower the compute capability of each core, but not too much
- use a scalable interconnection network fabric on-chip (NoC)
 - to minimize shared-cache/memory access latencies
 - to provide enough data communication bandwidth
 - to minimize traffic bottlenecks
- group cores in clusters to improve the quality of the NoC
- reduced cache size/levels (*beware: strong impact on performance...*)
- smaller fabrication process (Sapphire Rapids: 7nm; Zen 4: 5 nm; Apple M3: 3nm)
- mix general-purpose PUs with application-oriented modules:
GPUs for vector computing, TPU for tensor computing, ...
- move to MCM/chiplets (*simpler chips have better wafer production yield*)

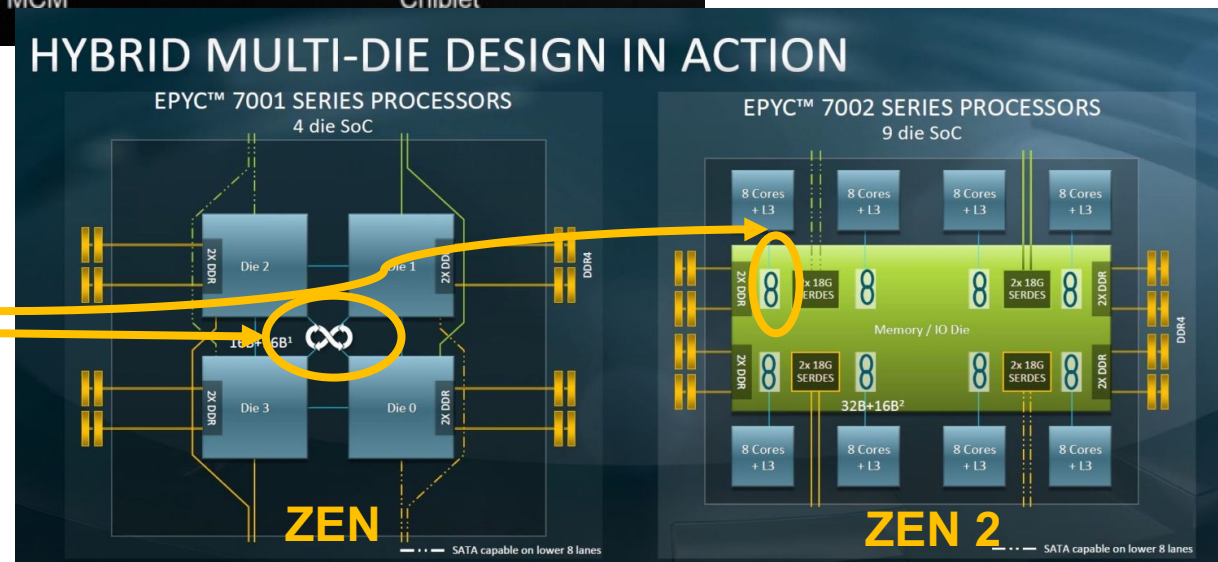


Key Intel Xeon competitor: AMD Epyc (Zen, Zen 2, 3, 4)

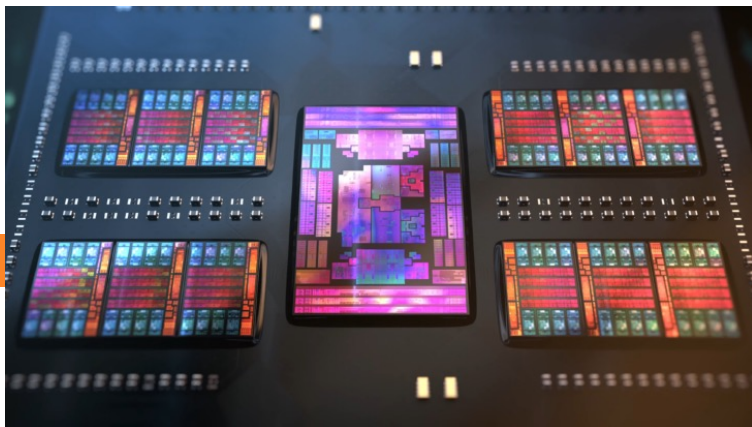


Infinity:

AMD interconnection fabric,
a superset of HyperTransport



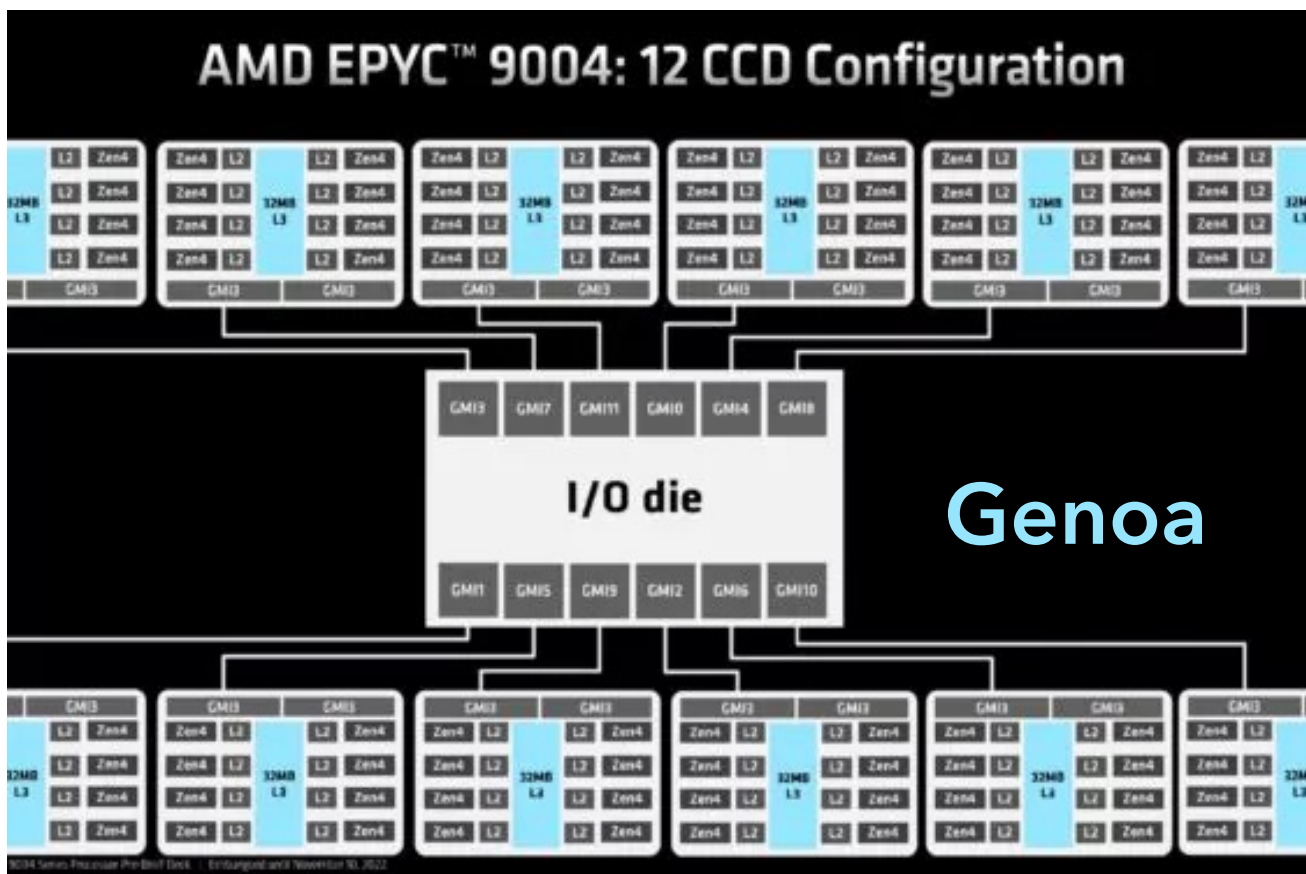




AMD 4th Gen Epyc – Zen 4 (Genoa & Bergamo)

Genoa: up to **5nm 96 cores** /192 threads
BFLOAT16, VNNU, AVX-512

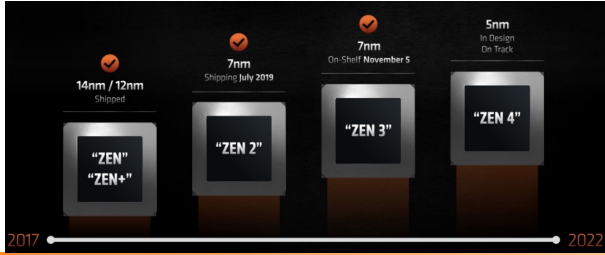
1 MiB/core L2 cache
384 MiB L3 cache
12 **DDR5** mem chan
128 PCIe Gen 5.0
(launched Nov'22)



Bergamo:

- up to **128 Zen 4c** cores
- for cloud-native computing
(expected 2H 2023)

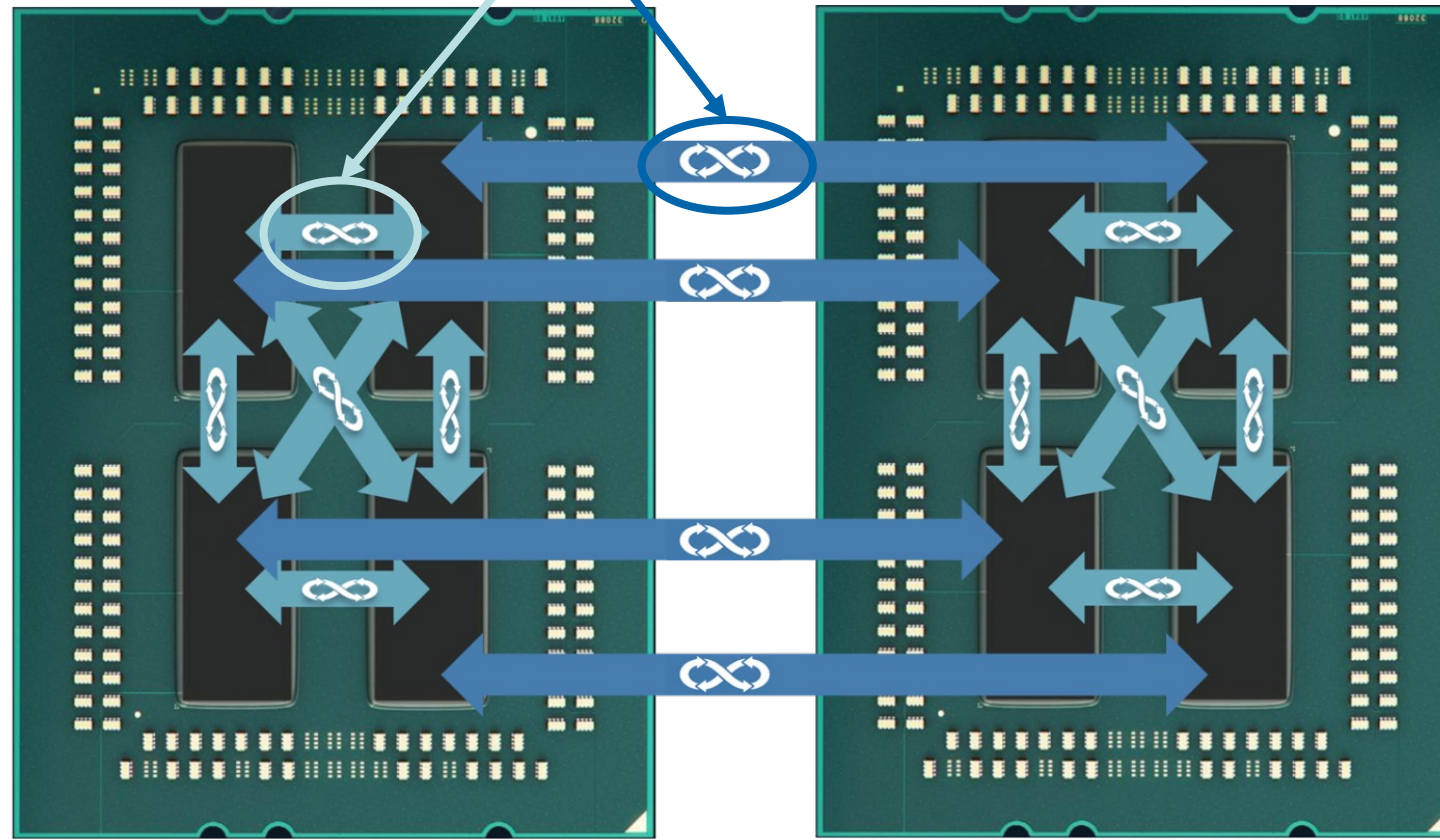
<https://wccftch.com/amd-4th-gen-epyc-genoa-zen-4-cpus-official-launch-96-cores-192-threads-worlds-fastest-server-chips/>



The AMD Infinity Fabric

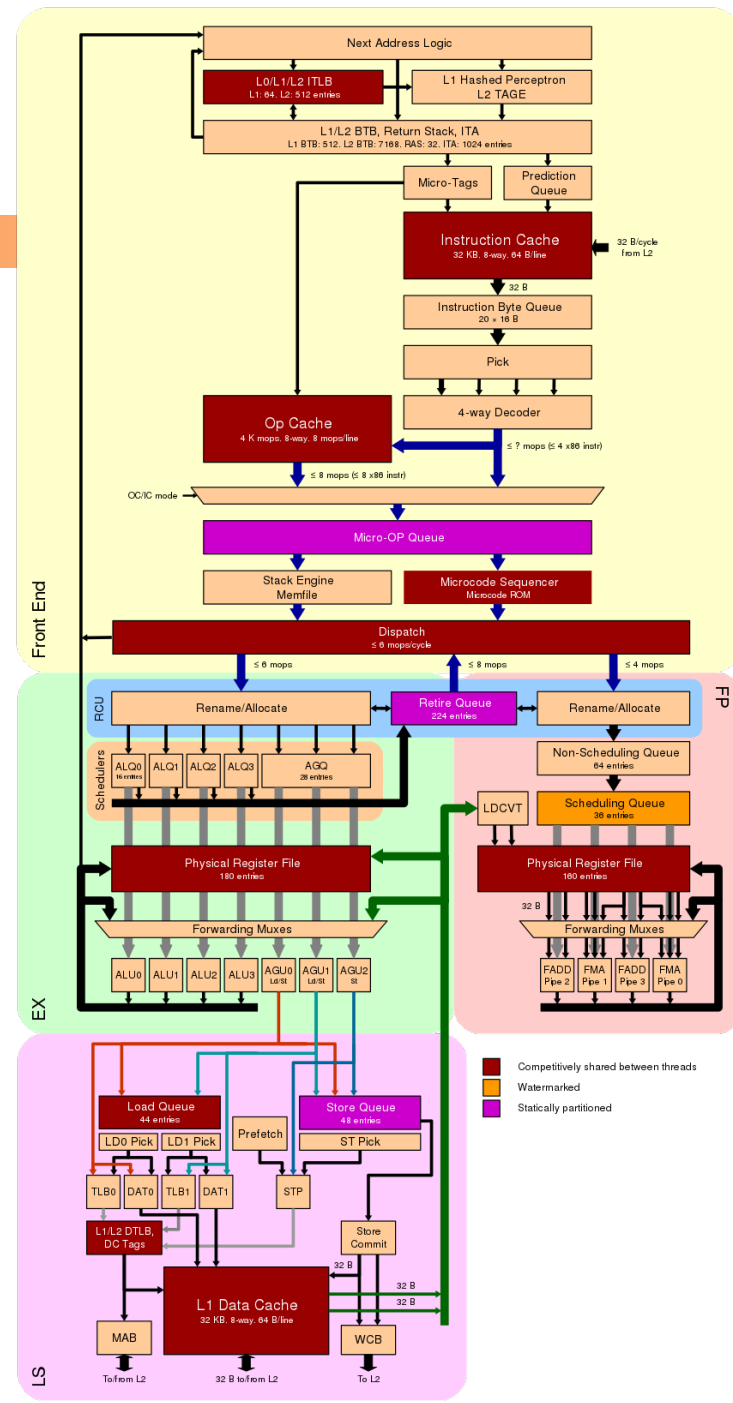
Infinity Fabric (IF)

- AMD system interconnect architecture, a 256-wide bi-directional crossbar:
 - **Infinity Fabric On-Package (IFOP)**: die-to-die communication in same package
 - **Infinity Fabric InterSocket (IFIS)**: for package-to-package communications



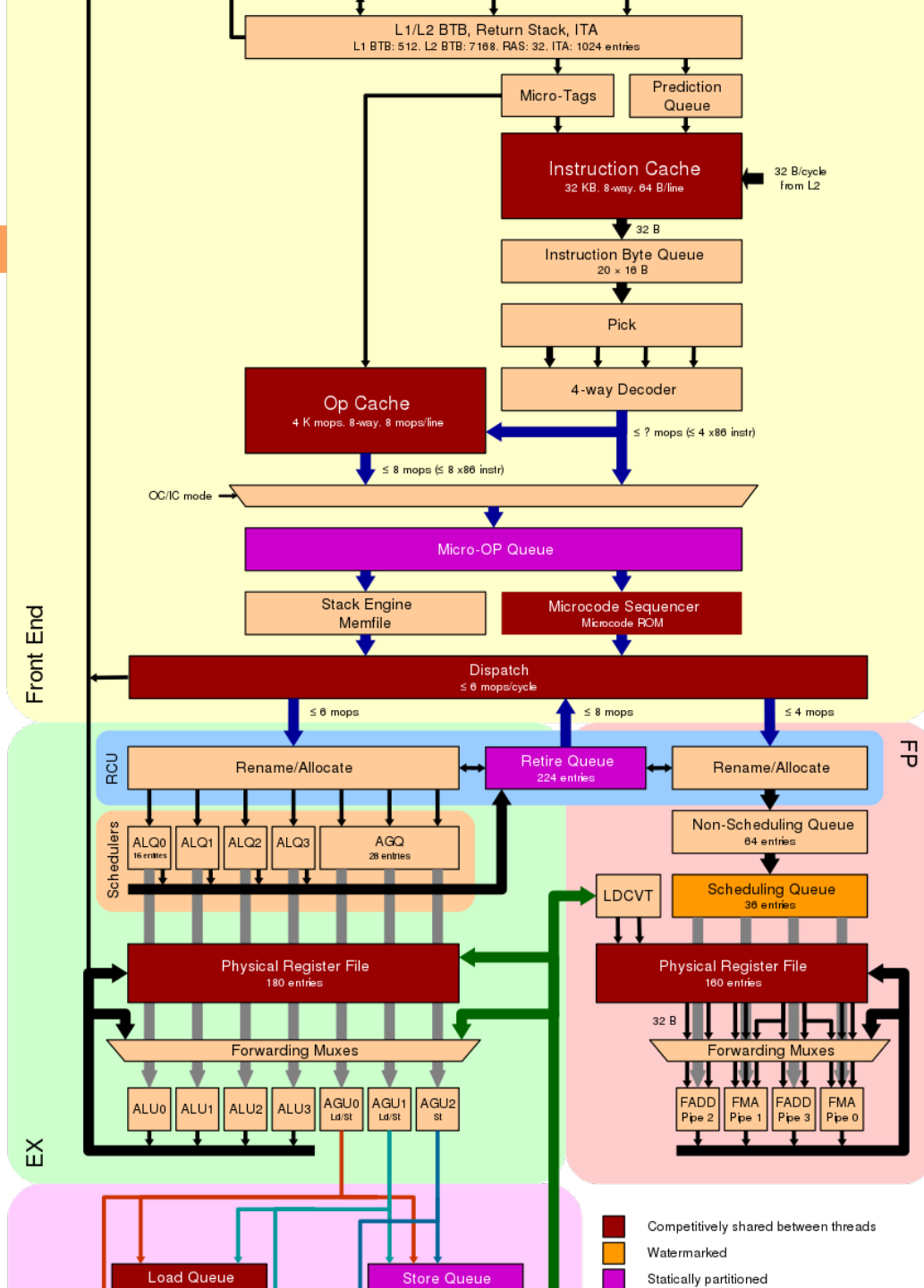


Zen 2 Block diagram

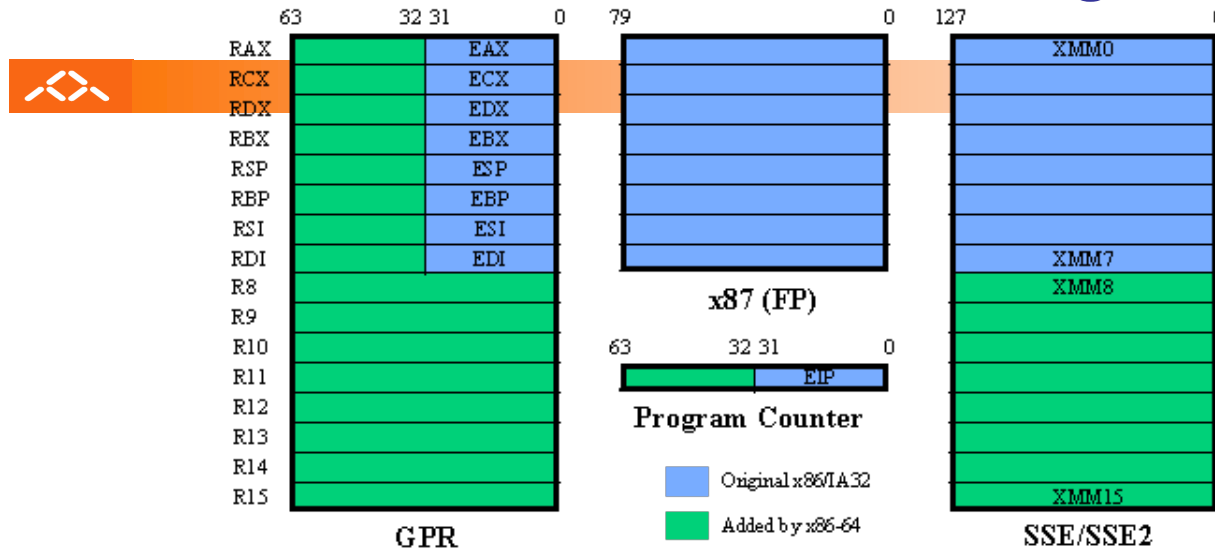




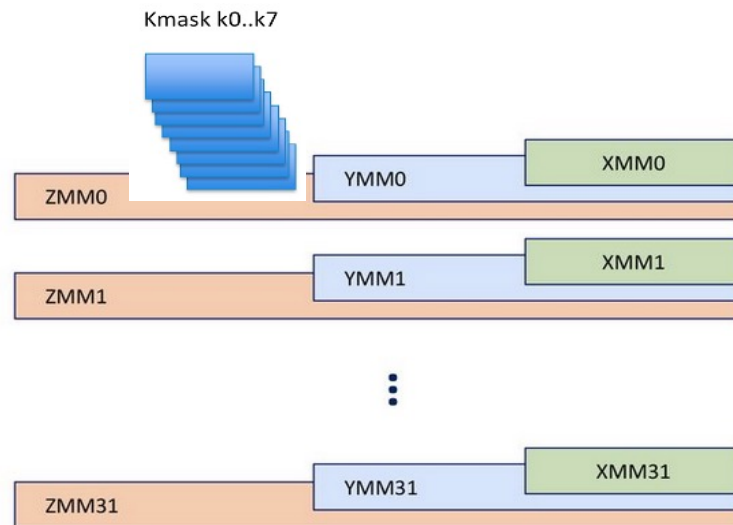
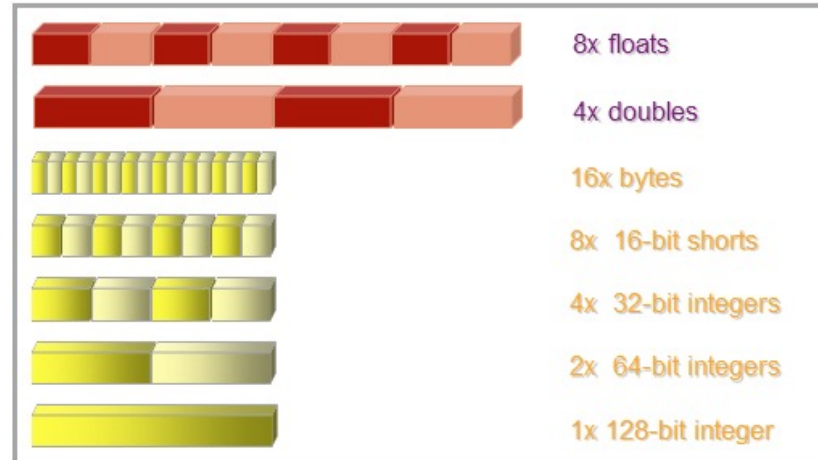
Zen 2 Block diagram



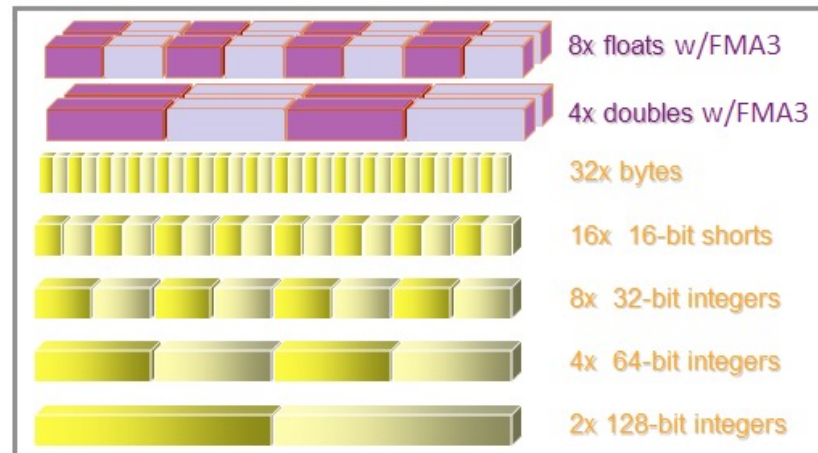
Registers for vector processing in Intel 64



Advanced Vector eXtensions, AVX 1.0



- ☐ XMM0 – XMM15
 - 128-bit registers
 - SSE
- ☐ YMM0 – YMM15
 - 256-bit registers
 - AVX, AVX2
- ☐ ZMM0 – ZMM31
 - 512-bit registers
 - AVX-512



AVX 2.0

AMD Zen 4 supports AVX-512, but...

The AVX-512 across Intel devices



Microarchitecture	Support Level													
	F	CD	ER	PF	BW	DQ	VL	IFMA	VBMI	4FMAPS	VNNI	4VNNIW	VPOPCNTDQ	BF16
Knights Landing	✓	✓	✓	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
Knights Mill	✓	✓	✓	✓	✗	✗	✗	✗	✗	✓	✓	✓	✓	✗
Skylake (server)	✓	✓	✗	✗	✓	✓	✓	✗	✗	✗	✗	✗	✗	✗
Cascade Lake	✓	✓	✗	✗	✓	✓	✓	✗	✗	✗	✓	✗	✗	✗
Cannon Lake	✓	✓	✗	✗	✓	✓	✓	✓	✓	✗		✗	✗	✗
Ice Lake (server)	✓	✓	✗	✗	✓	✓	✓	✓	✓	✓	✓	✗	✗	✗
Cooper Lake	✓	✓	✗	✗	✓	✓	✓	✗	✗	✗	✓	✗	✗	✓

AVX512F - [AVX-512 Foundation](#)

AVX512CD - [AVX-512 Conflict Detection](#)

AVX512BW - [AVX-512 Byte and Word](#)

AVX512DQ - [AVX-512 Doubleword and Quadword](#)

AVX512VL - [AVX-512 Vector Length](#)

AVX512IFMA - [AVX-512 Integer Fused Multiply-Add](#)

AVX512_VNNI - [AVX-512 Vector Neural Network Instructions](#)

AVX512_BF16 - [AVX-512 BFloat16 Instructions](#)

*"I hope AVX512 dies a painful death,
and that Intel starts
fixing real problems..."*
Linus Torvalds

HPCs with ARMv8 – server-level competitors (back in 2019...)

<https://www.eweek.com/pc-hardware/eweek-s-top-vendors-server-processor-makers>



1. Marvell ThunderX product family (**dead?...**)



2. Fujitsu A64FX Arm chip (**in MACC...**)



3. Neoverse N1 hyperscale reference design (...)



4. Ampere Altra Arm Processor (...)



5. Amazon Graviton (...)



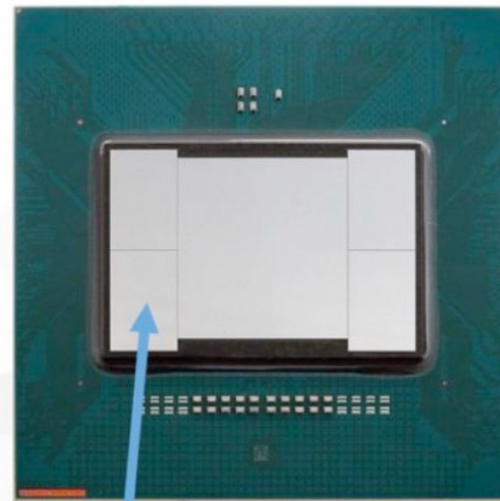
6. Huawei HiSilicon Kunpeng 920 (**dead?...**)



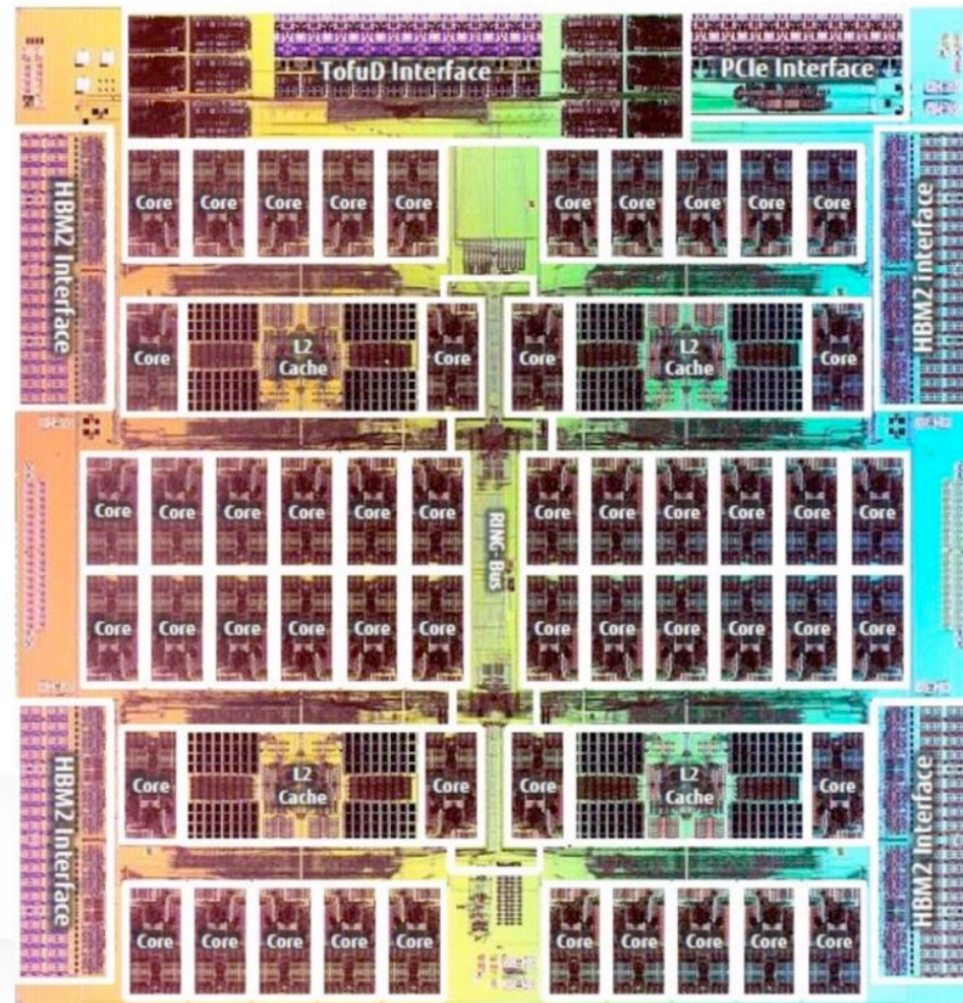
Fujitsu's A64FX ARM Chip (#1 in June'20 TOP500)



- TSMC 7nm FinFET
- CoWoS technologies for HBM2



HBM2



<https://www.scientific-computing.com/sites/default/files/content/white-paper/pdfs/EXDCI-2-White%20paper%20Future%20HPC%20technologies.pdf>



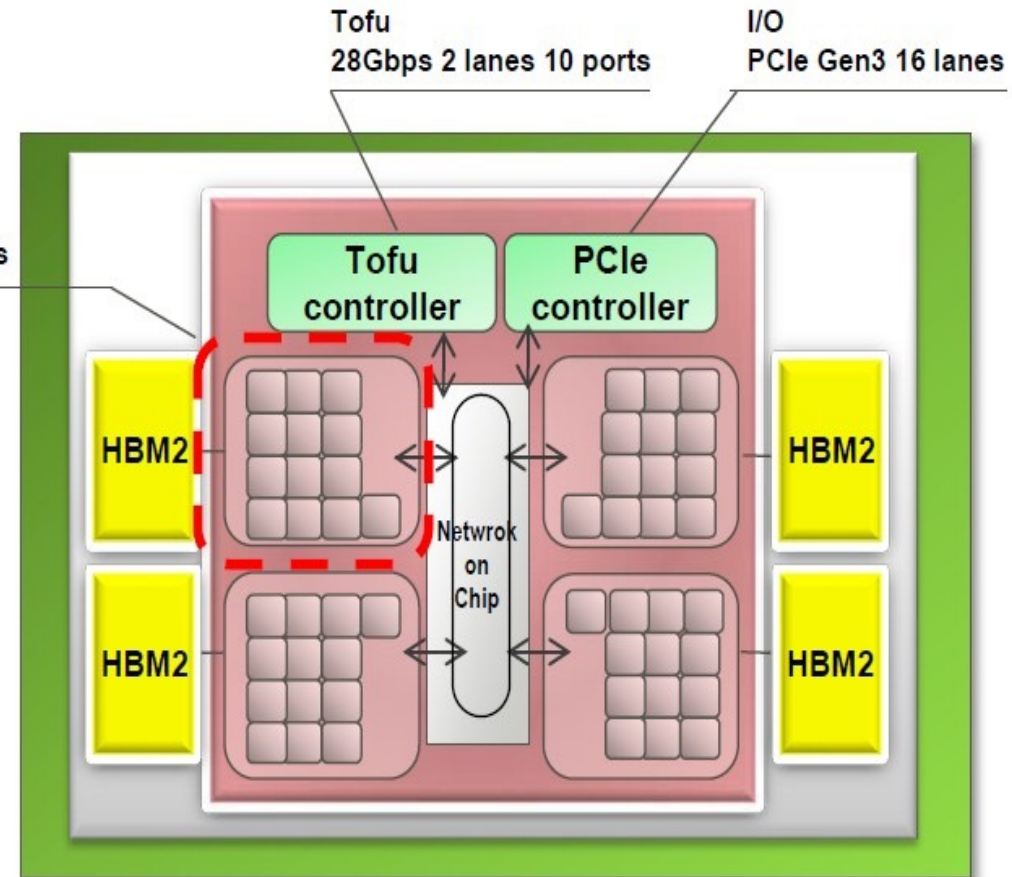
Fujitsu's A64FX Arm Chip – 48+4 cores



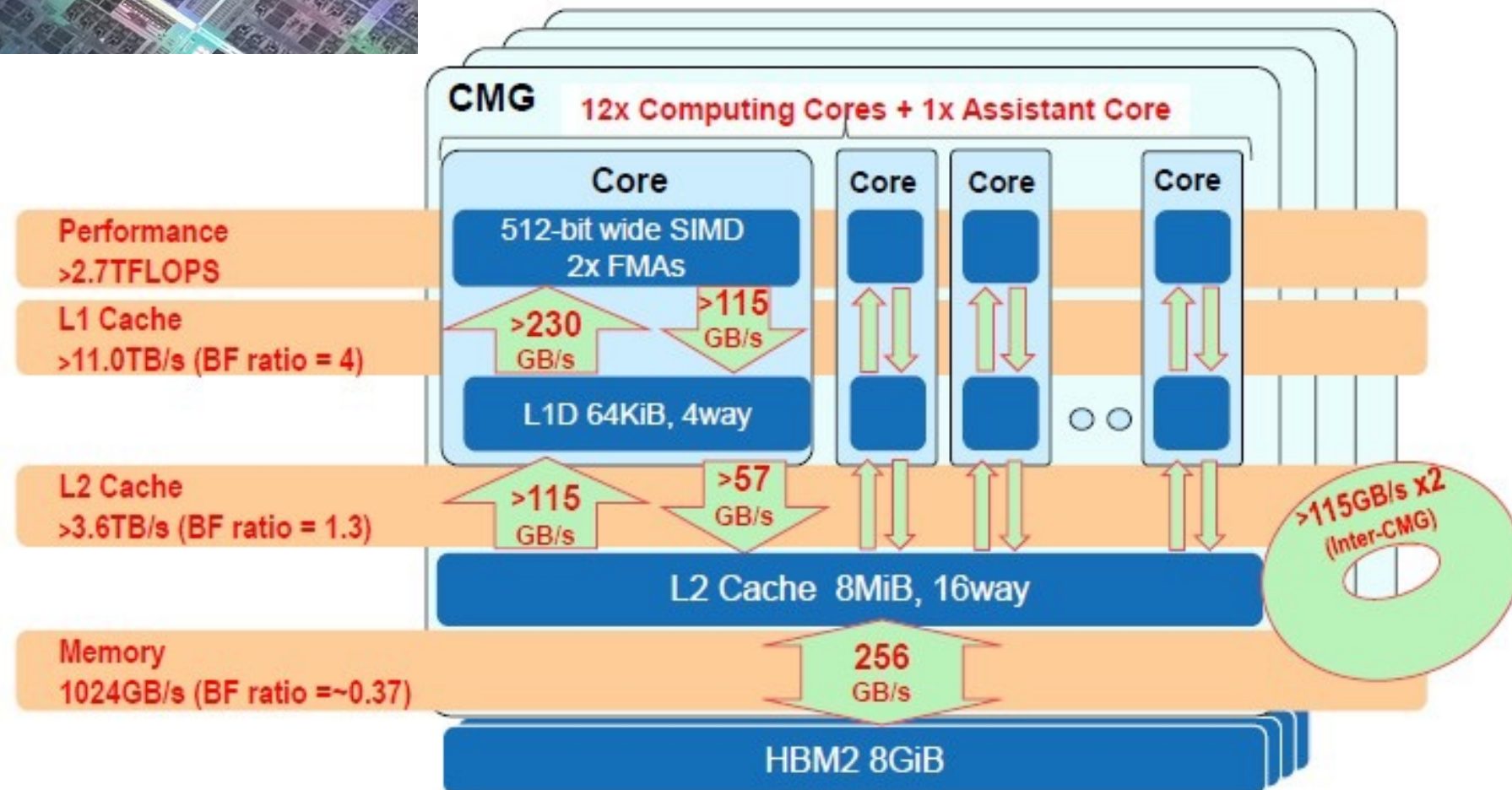
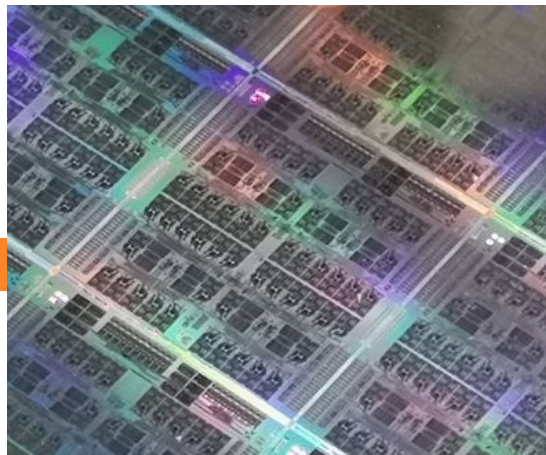
A64FX Arm

- Feature size: 7 nm
- Armv8.2-A spec with 512-bit SVE extensions
- HP math and a dot-product engine
- 4 core-memory groups
- NoC: a double ring bus
- cores in CMG linked by a crossbar to L2 cache & to HBM2 mem controller
- 8 MiB L2 cache; no L3 cache
- a Tofu-D controller on the die
- #1 TOP500 from Jun'20 to Nov'21 uses A64FX package

CMG specification
13 cores
L2\$ 8MiB
Mem 8GiB, 256GB/s



Block diagram of the A64FX chip

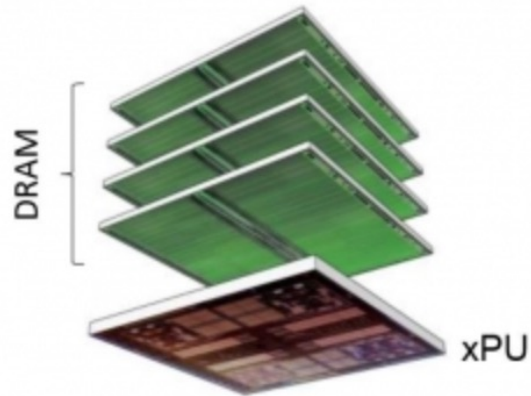


3D Stacked DRAM

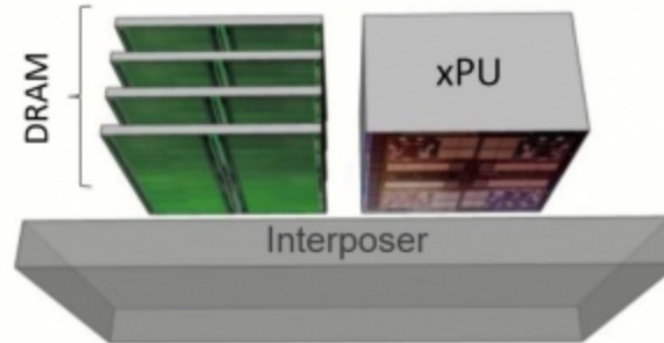


BUT FIRST... THERE ARE TWO FLAVORS OF DIE STACKING

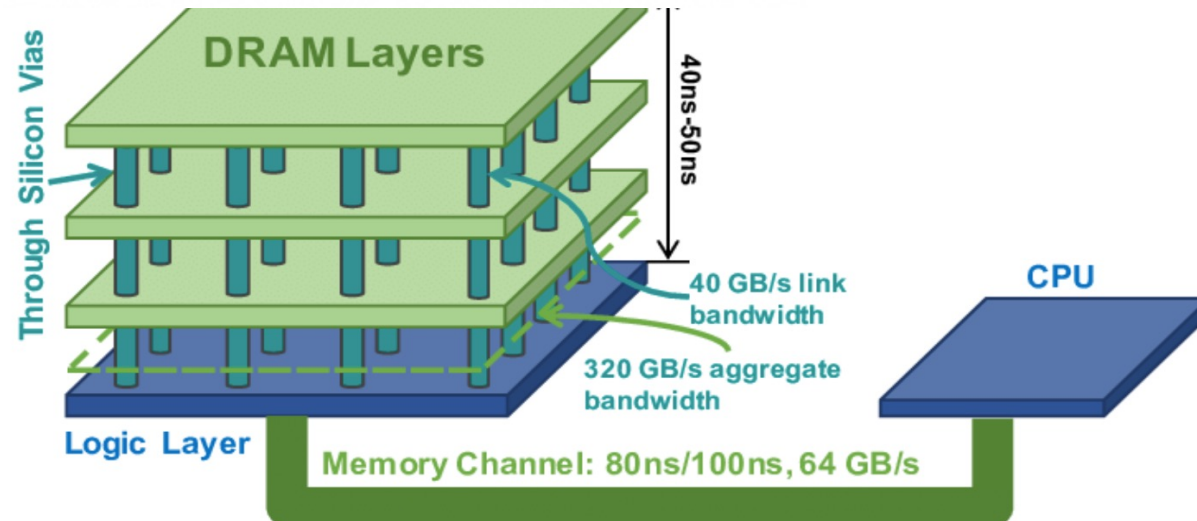
AMD



VERTICAL STACKING (3D)



INTERPOSER STACKING (2.5D)



NEON vector & FP registers in Armv8 (64-bit)

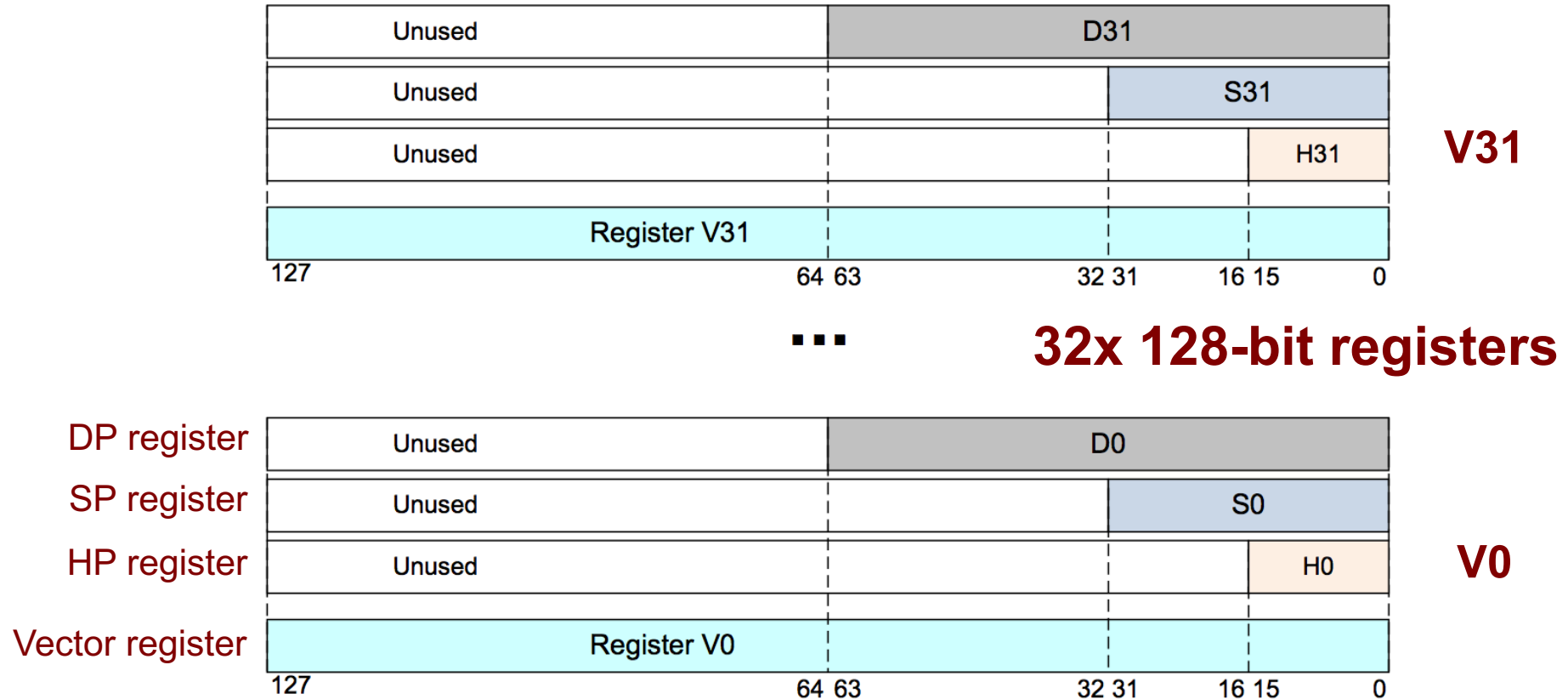


Figure 4-10 Arrangement of floating-point values

Note

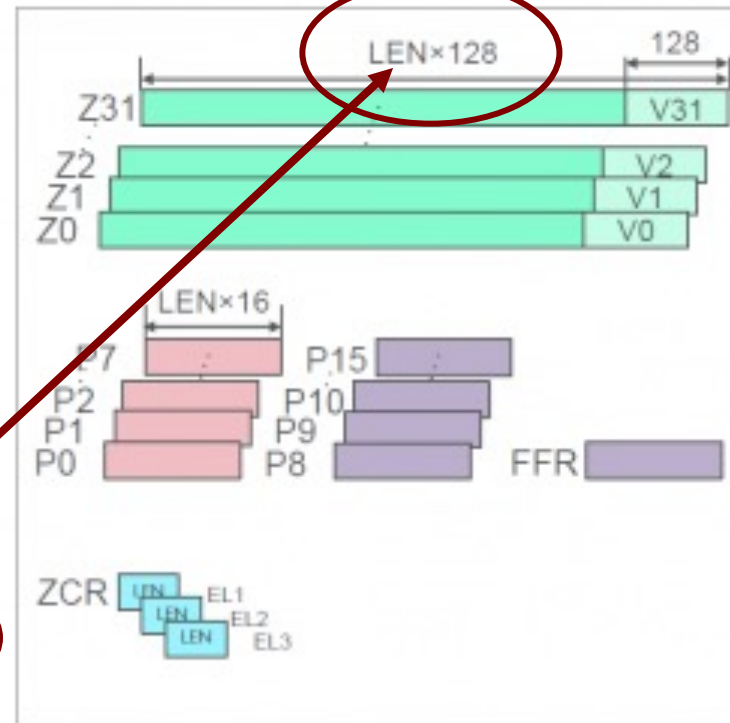
16-bit floating-point is supported, but only as a format to be converted from or to. It is not supported for data processing operations.

Armv8-A Scalable Vector Extension (SVE)



SVE architectural state

- Scalable vector registers
 - Z0-Z31 extending NEON's V0-V31
 - DP & SP floating-point
 - 64, 32, 16 & 8-bit integer
- Scalable predicate registers
 - P0-P7 lane masks for ld/st/arith
 - P8-P15 for predicate manipulation
 - FFR *first fault register*
- Scalable vector control registers
 - ZCR_ELx vector length (LEN=1..16)
 - Exception / privilege level EL1 to EL3



Vector size: up to **2048**